Lung Nodule Detection on the LUNA16 Dataset

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I. Introduction

Lung cancer remains one of the leading causes of cancer-related mortality worldwide. Early detection of pulmonary nodules in chest CT scans is essential for improving patient outcomes, yet manual review by radiologists is both time-consuming and error-prone. Deep learning-based computer-aided detection systems offer a promising path toward automating this process. This report describes an experiment applying a 3D convolutional neural network (CNN) to the LUNA16 benchmark for nodule candidate classification. Due to Kaggle platform limitations, the experiment operates on a restricted subset of the data, covering the full pipeline from raw CT loading to evaluation.

II. Dataset

LUNA16 [1] is derived from the LIDC-IDRI collection and contains 888 chest CT scans with consensus nodule annotations from multiple radiologists, alongside a candidates file of labeled locations (nodule vs. non-nodule). Scans are split into ten subsets for cross-validation.

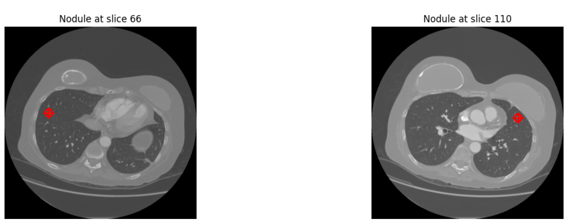

Fig. 1. Sample data with nodule

Due to memory and storage constraints on the free Kaggle tier, this experiment uses only **subset0**, limited to **50 scans**. After filtering, the working set contains a small number of positive (nodule) candidates and a much larger number of negatives, reflecting a strong class imbalance typical of the full dataset.

III. Methodology

A. Preprocessing

Each CT scan is loaded via SimpleITK. Candidate world coordinates are converted to voxel space using each scan's origin and spacing. A $32 \times 32 \times 32$ voxel patch is extracted around each candidate, with zero-padding applied at boundaries. Intensities are clipped to the range $[-1000, 400]$ Hounsfield Units and rescaled to $[0, 1]$ to isolate relevant lung tissue contrast.

B. Model Architecture

The model is a 3D CNN taking $32 \times 32 \times 32 \times 1$ patches as input. It consists of three convolutional blocks, each with a $3 \times 3 \times 3$ convolution (ReLU), 3D max pooling, and batch normalization, with filter counts of 32, 64, and 128. A global average pooling layer is followed by a 128-unit dense layer, dropout (rate 0.5), and a sigmoid output neuron.

C. Training Configuration

The model is compiled with the Adam optimizer (learning rate 10^{-3}) and binary cross-entropy loss. Training runs for up to 50 epochs with batch size 8. Early stopping (patience 10) and learning rate reduction on plateau (factor 0.5, patience 5) are used as callbacks. To address class imbalance, negative samples are undersampled to at most $10\times$ the positives, and the positive class is assigned a loss weight proportional to the negative-to-positive ratio.

D. Data Augmentation

No augmentation was applied. Techniques such as random flipping or rotation were omitted to avoid exceeding the Kaggle runtime memory budget.

IV. Results

A. Overall Performance

TABLE I
Validation set performance.

Metric	Value
Accuracy	$\sim$0.75
Precision	$\sim$0.14
Recall	$\sim$0.43
F1-Score	$\sim$0.21
AUC-ROC	0.64

The model correctly classified 115 non-nodules and 6 nodules, with 30 false positives and 8 false negatives (Figure 2). While accuracy appears reasonable at 75%, it is inflated by the class imbalance. The low precision (0.14) shows that most positive predictions are incorrect, and a recall of 0.43 means over half of actual nodules are missed. The AUC of 0.64 (Figure 3) confirms only modest discriminative ability above random.

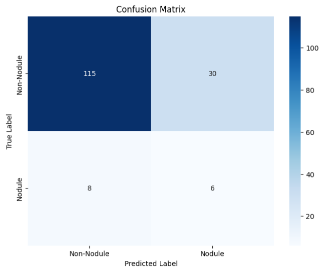

Fig. 2. Confusion matrix

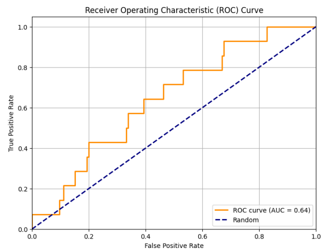

Fig. 3. ROC curve (AUC = 0.64)

B. Training Dynamics

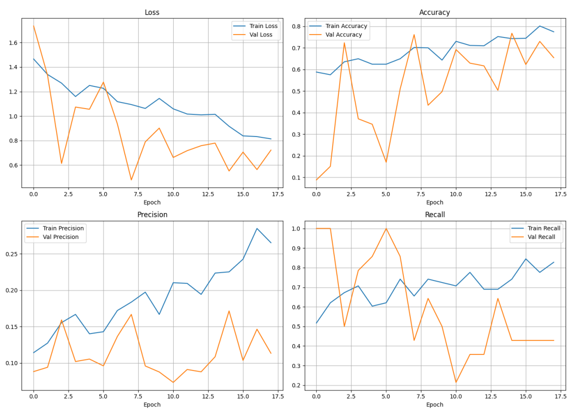

Fig. 4. Training and validation metrics

Training ran for 18 epochs before early stopping triggered. The training loss decreased steadily while validation loss oscillated with high variance, indicating instability driven by the small dataset and class imbalance. Validation precision remained consistently low throughout, never exceeding 0.17, while recall fluctuated between 0.2 and 1.0 across epochs.

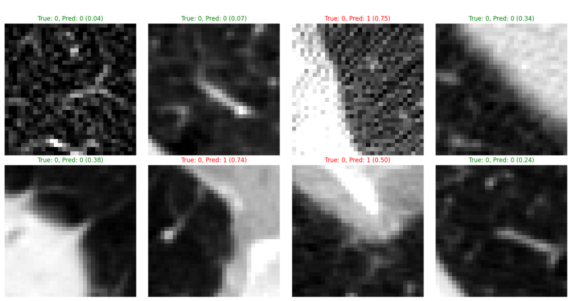

Fig. 5. Result Visualization

V. Discussion

A. Limitations

The most significant constraint is data scarcity — 50 scans from a single subset is far below what a 3D CNN typically needs to generalise. Class imbalance, even after undersampling and loss weighting, continues to suppress precision. The absence of augmentation further limits the model's ability to learn robust features. Additionally, patches were extracted without resampling to a common isotropic spacing, meaning nodules of different physical sizes appear at varying scales across scans.

B. Future Work

Using all ten subsets with proper 10-fold cross-validation would be the most impactful improvement. Adding standard augmentations (flips, rotations, intensity jitter) and resampling scans to isotropic $1\,\mathrm{mm}^3$ spacing would also help. More advanced architectures such as residual 3D networks or attention-based models could further improve performance. Finally, tuning the classification threshold beyond the default 0.5 to optimise F1-score would better reflect clinical priorities.

VI. Conclusion

This work implemented a complete 3D CNN pipeline for lung nodule detection on a constrained subset of LUNA16. The model achieved an AUC of 0.64 and recall of approximately 0.43, demonstrating that some discriminative signal was learned despite limited data. The results highlight the critical role that dataset size, class balance, and augmentation play in medical image classification tasks. Addressing these constraints in future iterations would be the clearest path toward clinically meaningful performance.

References

[1] A. A. A. Setio et al., "Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge," *Medical Image Analysis*, vol. 42, pp. 1–13, 2017.

[2] AVC0706, "LUNA16 dataset," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/avc0706/luna16